Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 7 · day per timetable	7	1 period (60 min)	Overview of Complex Numbers	Algebra II	Topic 2 — Quadratic Functions and Equations · Lesson 2-4	Grade 10 — 10A and 10C

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Canva (final product) · Squared paper	Lesson presentation (15 slides) · "Choose Your Route" differentiation sheet · "NEOM Pump Circuit" project task (A3) · "Give it a go" classwork

Competencies	Technology Embedded	Connection to student's real life
1- Simplifying square roots of negative numbers, evaluating powers of i, and adding, subtracting and multiplying complex numbers. 2- Dividing complex numbers with the conjugate and writing the solutions of a quadratic equation in the form $a+bi$.	1- Quizizz live diagnostic with instant gap analysis. 2- Calculator check: confirm the modulus with the square-root key, rounding only at the last step.	1- A series circuit at a coastal desalination plant: the total impedance, the current and the breaker decision. 2- Engineers who use complex numbers to describe alternating current.

Values	Soft Skills	Hard Skills
1- Precision — replacing i^2 by -1 every time. 2- Safety — checking a result against a limit before deciding.	1- Collaboration in assigned group roles. 2- Explaining in words why the conjugate removes i from the denominator.	1- Operations with complex numbers in the form $a+bi$. 2- Solving quadratic equations with complex solutions.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Demonstrate the role of the imaginary unit i in the system of complex numbers. 2- Evaluate expressions involving the imaginary unit i and complex numbers. 3- Use the properties of complex numbers to perform operations such as addition, subtraction, and multiplication. 4- Use complex conjugates to simplify quotients of complex numbers. 5- Solve real-world problems using complex conjugates. 6- Express the complex solutions of a quadratic equation in the form a + bi. *Vocabulary: 12 terms — listed in the FIKR plan.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: $\sqrt{72}$, $(x+3)(x-2)$, the discriminant of x^2+2x+5. Results read as a gap map; students routed to the re-teach table, straight to i, or the Investigate prompt. FIKR Phase 2 — Targeted Instruction. Whole class: the imaginary unit; adding, subtracting and multiplying; conjugates and modulus; division and complex solutions. Quick check on whiteboards. FIKR Phase 3 — Practice. Two identical "Give it a go" problems worked together. Students then choose a route and may switch at any time. Live feedback via Pear Deck; 2-minute micro-conferences. FIKR Phase 4 — Production. The NEOM series circuit: total impedance, the current in the form $a+bi$, its modulus and the breaker decision. AI used as critic only. Extended in the A3 project task. FIKR Phase 5 — Proof of Learning. "Time to Check" — a power of i, a product and a quotient. No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product including one worked non-example, and publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell on the final products, then the reflection question: "Which is harder for you — remembering that $i^2=-1$ inside a product, or dividing with the conjugate?"
Reflection for Improvement:	
Homework:	Finish the final product and upload to the LMS portfolio (any of the three formats). Practice set on Pear Deck — 12 questions, choose your route. Watch the flipped video for the next lesson, Topic 2 · Lesson 2-5 — Completing the Square. Clinic group: bring your Time to Check paper.

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